



RAW DATA TO PRODUCTION DASHBOARDS

Activation Webinar Series

Account Engineering Team | September 2026

Access GitHub Repo

Clone or download the complete source code and scripts for this workshop



Raw Data to Production Dashboards Repo

Includes end-to-end implementation code for Bronze, Silver, and Gold dynamic tables, pipeline orchestration with Streams & Tasks, and Streamlit dashboard templates.

[Open Repository](#) →



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What We'll Build

One pipeline from raw orders to a refreshing revenue dashboard

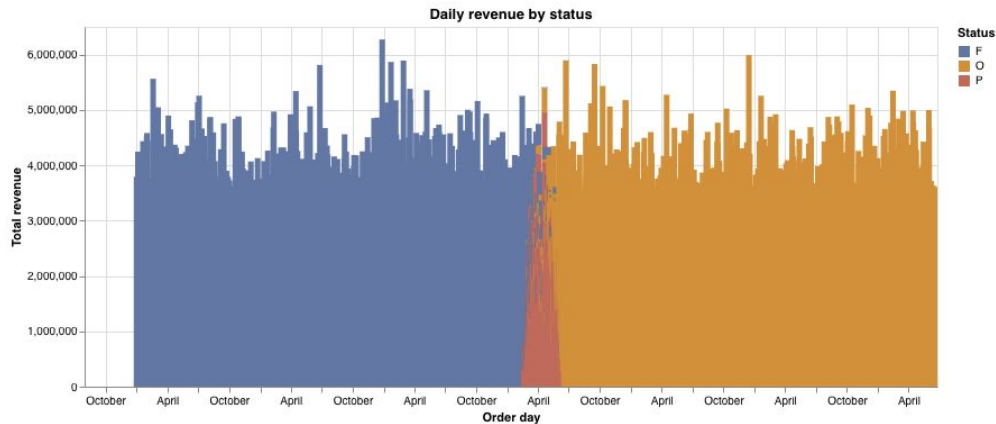
1 Land 50,000 raw orders in Bronze

2 Refine to Silver with a Dynamic Table

3 Aggregate to Gold for reporting

4 React to change with Streams and Tasks

5 Serve Gold to an approved dashboard



By the end, the Gold Dynamic Table refreshes against a lag target and the notebook dashboard reads straight from it.



SECTION 1

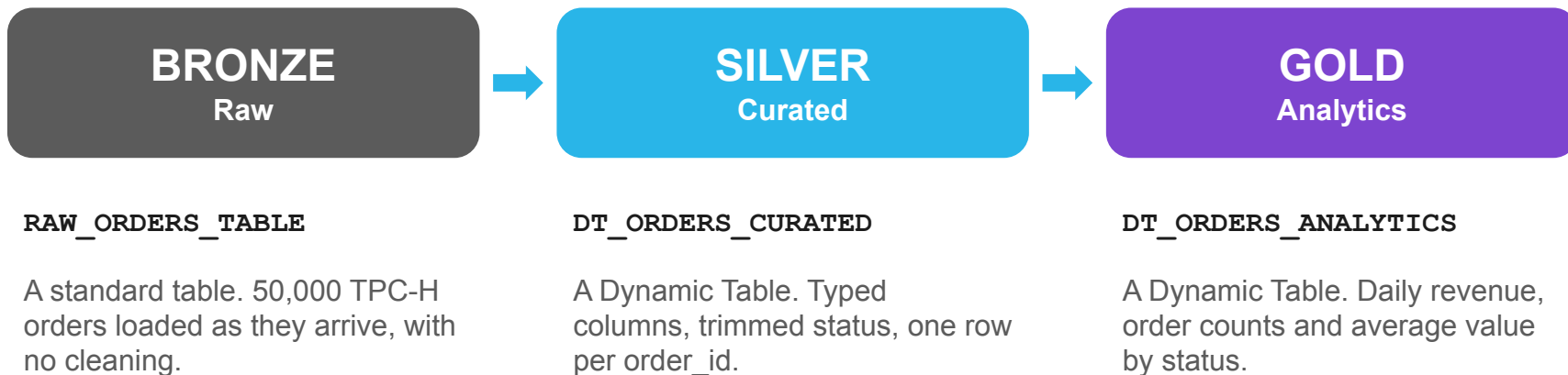
MEDALLION ARCHITECTURE

OVERVIEW (BRONZE / SILVER / GOLD)



Medallion Architecture

Three layers in one database, progressively refined

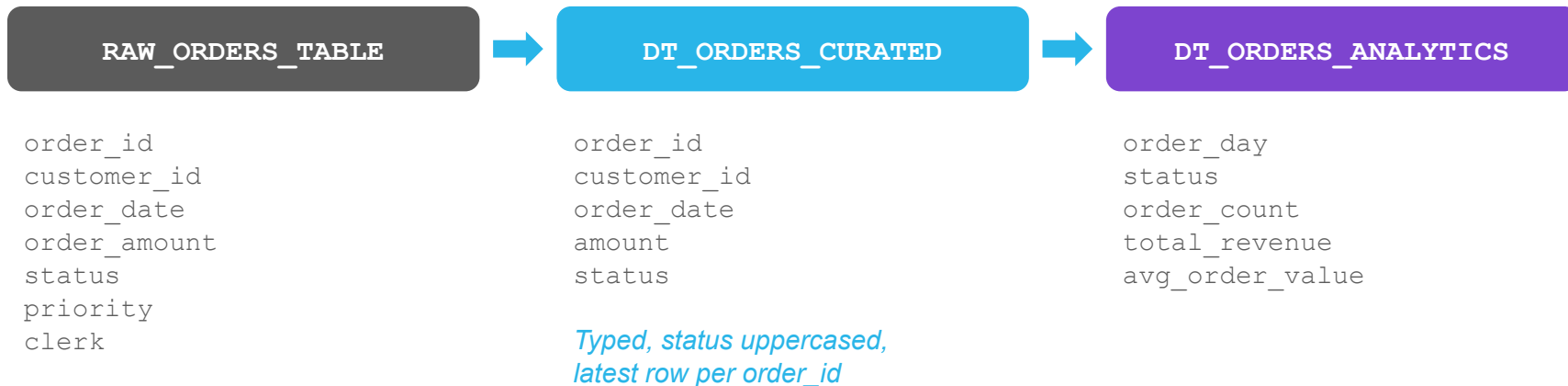


Bronze, Silver and Gold are naming conventions in this lab, not enforced guarantees. All three objects live in the same PUBLIC schema.



From Orders to Daily Revenue

What each layer actually contains in this lab



Silver keeps the order grain. The grain changes at Gold: one row per order becomes one row per day and status.



SECTION 2

DYNAMIC TABLES — DECLARATIVE PIPELINES



Dynamic Tables: Define the Result

You declare the output; Snowflake maintains it against a freshness target

Declare the Result

Set a Freshness Target

Define the Transformation

```
CREATE OR REPLACE DYNAMIC TABLE DT_ORDERS_CURATED
  TARGET_LAG = '1 minute'
  WAREHOUSE = RAW_DATA_TO_PRODUCTION_DASHBOARDS_WH
  REFRESH_MODE = INCREMENTAL
AS
SELECT order_id, customer_id, order_date,
       amount, status
FROM RAW_ORDERS_TABLE
WHERE order_id IS NOT NULL
QUALIFY ROW_NUMBER() OVER (
  PARTITION BY order_id ORDER BY order_date DESC
) = 1;
```

TARGET_LAG is a best-effort staleness target, not a refresh interval. REFRESH_MODE resolves at creation — confirm it with SHOW DYNAMIC TABLES.



Choose the Mode, Inspect the Refresh

Refresh mode resolves at creation — then you verify what it chose

INCREMENTAL

Applies only the changed rows.
Requires a supported query shape.

FULL

Recomputes the entire result
from scratch on every refresh.

ADAPTIVE

Normally refreshes
incrementally, and may
reinitialize.

```
SHOW DYNAMIC TABLES;  -- refresh_mode, refresh_mode_reason

SELECT name, refresh_action, reinit_reason
FROM TABLE(INFORMATION_SCHEMA.DYNAMIC_TABLE_REFRESH_HISTORY());
```

AUTO resolves once at creation and does not re-choose per refresh. In this lab the EXCEPT example resolves to FULL.



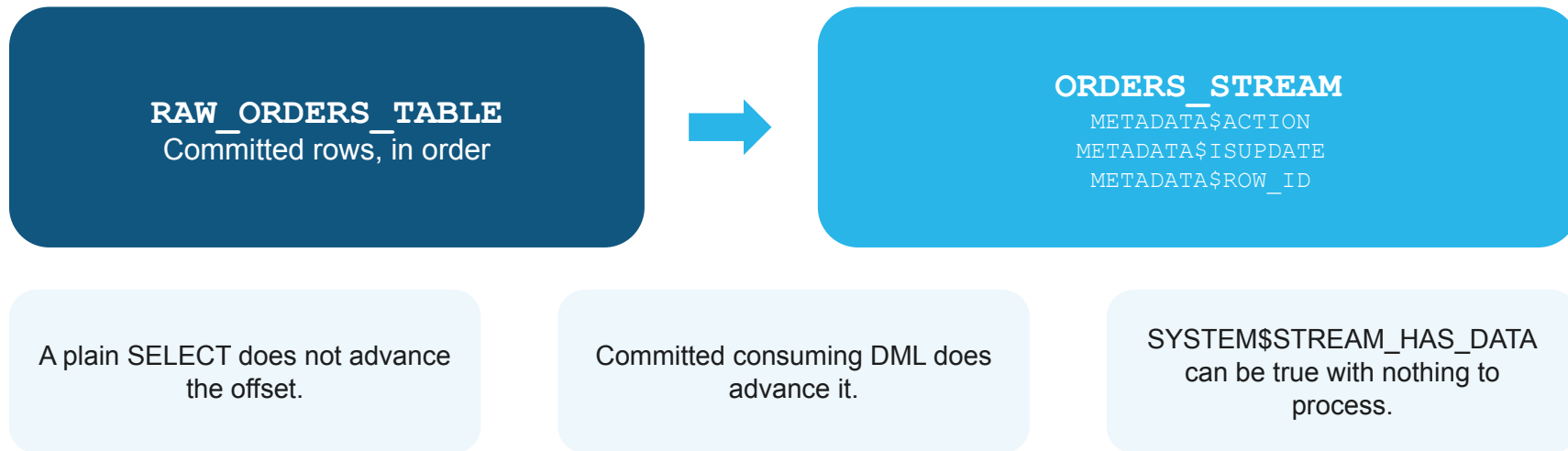
SECTION 3

STREAMS + TASKS — EVENT-DRIVEN CDC



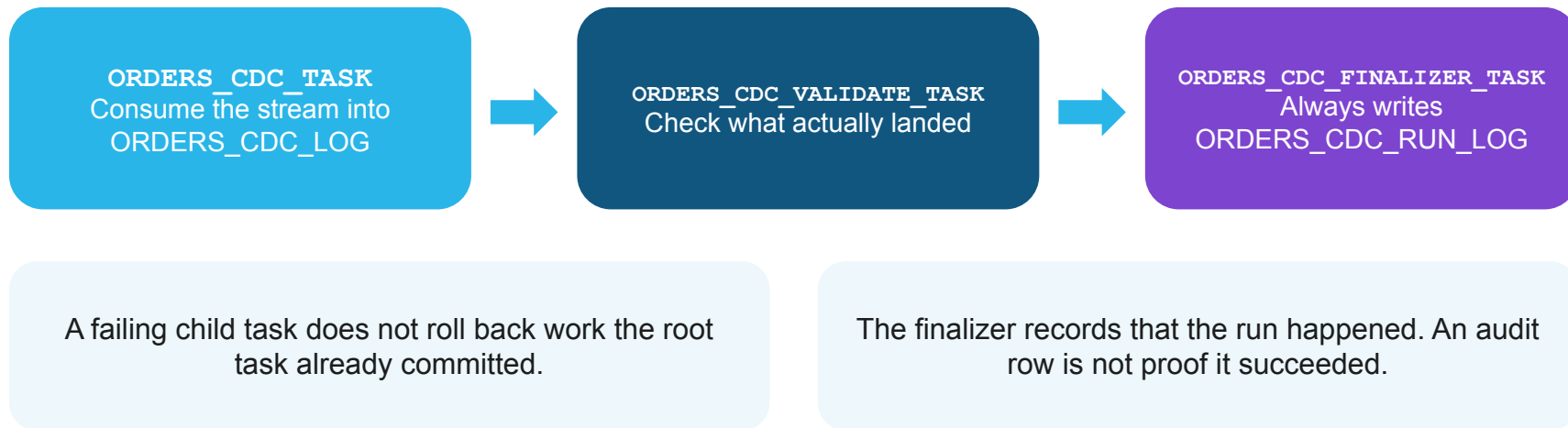
Table Stream: What Changed?

A stream tracks an offset, not a copy of your data



Capture, Validate, Finalize

A three-task graph, with a finalizer that always records the run



The runnable demo sends no email. The notification step is a walkthrough only.



SECTION 4

DBT ON SNOWFLAKE — CODE-FIRST TRANSFORMATIONS



dbt Models, Tests and Dependencies

Sources, models and tests resolve into a dependency graph

```
dbt_project.yml
```

```
models/  
  sources.yml      -- raw.RAW_ORDERS_TABLE  
  orders_clean.sql -- materialized='table'  
  schema.yml       -- not_null, unique
```

source: raw.RAW_ORDERS_TABLE



model: orders_clean



tests: not_null, unique

dbt build runs the selected tests; dbt run does not. This model and the Dynamic Table example both read RAW_ORDERS_TABLE, but they do not produce identical results.



From Repository to Execution

A git push is not a deployment — these are three distinct steps



A git push updates the repository. It does not deploy the project, and it does not run it.

Each step is a distinct object: a GIT REPOSITORY, a DBT PROJECT, and an execution or a TASK that runs it.



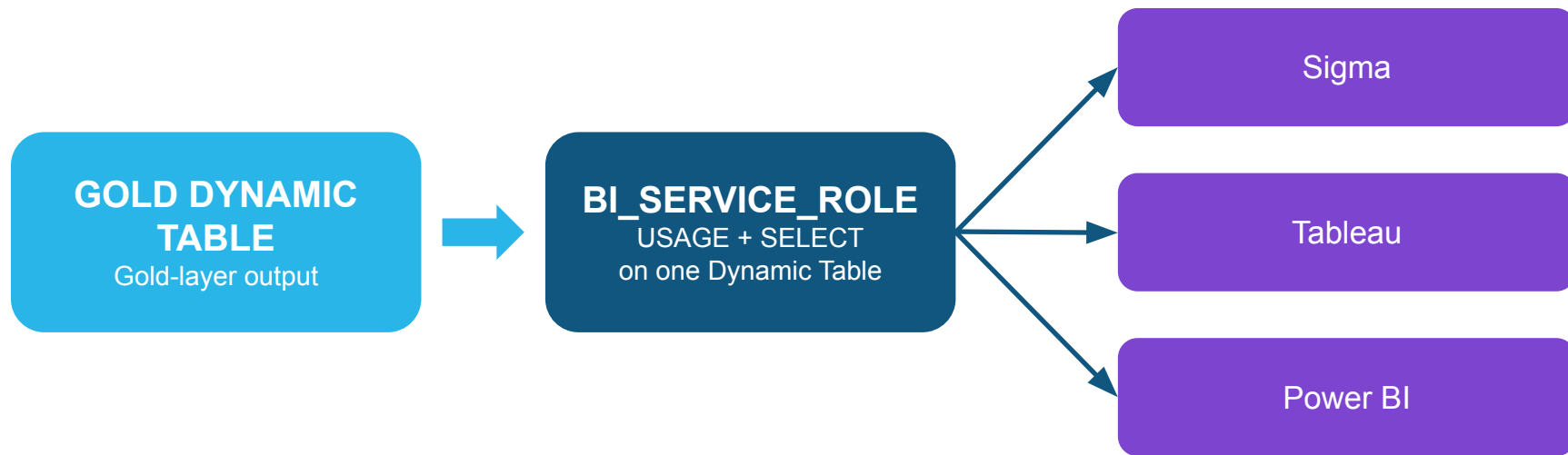
SECTION 5

BI TOOL CONNECTION + WORKSPACES DASHBOARD AND STREAMLIT PATH



Serve Gold Safely

One least-privilege role exposes the Gold Dynamic Table to approved BI tools



Read the Dashboard, Trace the Result

Every number on screen traces back to the Gold Dynamic Table

Gold rows = COUNT(*)

Total orders = SUM (ORDER_COUNT)

Total revenue = SUM (TOTAL_REVENUE)

Freshest order day = MAX (ORDER_DAY)

	<u>A</u> Gold rows	<u>A</u> Total orders	<u>A</u> Total revenue	<u>A</u> Freshest order day
1	2,594	50,000	\$7,650,456,966.13	1998-08-02

Freshest order day is the newest order date in the data, not the time the Dynamic Table last refreshed.



Choose the Control You Need

Three ways to transform, and what each one gives you

DYNAMIC TABLES

Declarative

You state the result. Snowflake maintains it against a lag target.

Best for keeping a table fresh.

STREAMS + TASKS

Imperative, event-driven

You control each step and can audit every run.

Best for custom logic on change.

dbt PROJECTS

Code-first

Version control, tests and a dependency graph.

Best for team-owned transformation projects.

These are not mutually exclusive. This lab uses all three against the same Bronze table.



THANK YOU



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